Beyond Spatial Accuracy: Diagnosing Persistent Orientation Failures in Vision–Language Models

Anonymous WACV Datasets Track submission

Paper ID *****

Abstract

Evaluating AI/ML systems responsibly requires practices for testing, stress-testing, auditing, comparing, and interpreting specific capabilities—not only aggregate benchmark scores. This paper proposes such a practice for spatial reasoning in vision–language models (VLMs): a diagnostic audit of object-relative orientation, a capability that aggregate accuracy can hide. On Visual Spatial Reasoning (VSR), moving from a compact 2.2B model to a larger 7B VLM substantially improves overall, depth, horizontal, and containment accuracy, but orientation changes only modestly; a large 2026 sparse-MoE model (MiMo-V2.5, 310B total / 15B active) reproduces the pattern (79.5% overall, 65.7% orientation). Except for structured prompting, which degrades performance, trained conditions on both models and the large sparse-MoE zero-shot model remain in a narrow 62–66% orientation band. We test structured prompting, general and targeted LoRA, hard negatives, visual/projector adaptation, frozen and object-grounded representation probes, and object-centric decomposition; none of the tested configurations reliably closes the gap, and a label-sensitivity analysis (an automated audit followed by two blind human re-audits; the humans agree well on the failure taxonomy, while clean/ambiguous exclusion masks remain annotator-dependent and automated auditors vary substantially in calibration) shows that the pattern persists after excluding examples flagged as ambiguous.

Accuracy also fails to capture relational coherence. LM-side adaptation significantly improves consistency over zero-shot, while hard-negative training raises facing/facing-away consistency from 66.0% to 77.7% with facing accuracy unchanged at 68.9%; the additional pooled consistency gain over LM-only is not significant. Cross-dataset evaluation on SITE shows that VSR-trained adaptation does not transfer cleanly: it is neutral overall and significantly harms official spatial-relationship reasoning (75.1%→71.6%, $p = 0.004$). A pre-specified orientation-vocabulary slice, frozen before SITE evaluation, is analyzed as exploratory evidence rather than as direct replication of the VSR construct. The results point to a persistent orientation bottleneck in which orientation is not robustly decodable by the tested frozen-feature readouts and relational decisions are not always propagated coherently, while also showing that task-specific spatial adaptation can transfer poorly. Each component of the audit—the diagnostic ladder, the label-sensitivity check, the accuracy–coherence separation, and the cross-dataset transfer protocol—is specified operationally so that it can be applied by others to other capabilities, benchmarks, and model families.

1. Introduction

This paper contributes an evaluation practice as much as a set of model results. We develop a protocol for testing, stress-testing, auditing, comparing, and interpreting one capability across vision–language models (VLMs), and instantiate it for object-relative orientation: facing, facing away, parallel, and perpendicular. This capability is relevant to robotics, embodied agents, navigation, and scene understanding, but it is easy to hide behind an aggregate score: success at identifying the relevant objects, inferring the relation, or answering complementary formulations does not guarantee success at the others.

Prior benchmarks establish that grounded spatial reasoning remains difficult, but they leave open how to diagnose a relation-specific failure. VSR introduced a controlled natural-image probe and reported particularly poor object-orientation performance [7]; What'sUp and GSR-Bench similarly expose failures on basic grounded relations [6, 9]; and broader evaluations continue to identify orientation as a difficult factor [15].

We therefore ask three narrower questions: does the VSR orientation weakness persist in modern generative VLMs and at much larger scale; can prompting, adaptation, or visual readouts remove it; and do accuracy, logical coherence, and cross-dataset transfer tell the same story?

The novelty is diagnostic rather than a claim that we discovered spatial difficulty. Our contribution is not any individual diagnostic, but a falsification protocol that holds the relation family and examples fixed while testing whether an apparent capability failure survives scaling and adaptation, label re-audit, representation readout, logical reformulation, and cross-dataset transfer. Prior work generally reports benchmark accuracy, studies a single intervention, or proposes a consistency objective; combining these views into one controlled audit of the same relation family is what lets us distinguish a persistent relation-specific bottleneck from aggregate underperformance, annotation artifacts, or benchmark-specific specialization.

In our VSR experiments, the distinction between relation families is striking. Moving from SmolVLM2-2.2B-Instruct to the larger Qwen2-VL-7B-Instruct increases zero-shot overall accuracy from 74.0% to 80.9%. Horizontal reasoning rises from 70.2% to 84.8%, depth from 68.9% to 75.2%, and containment from 83.4% to 89.3%. Orientation, by contrast, moves only from 62.8% to 63.5%. A large 2026 sparse-MoE model (MiMo-V2.5, 310B total / 15B active parameters) reproduces the same relation-specific pattern: 79.5% overall accuracy yet 65.7% on orientation, again inside the 62–66% band. We then evaluate interventions spanning prompting, language-side adaptation, targeted data construction, visual-side adaptation, representation probing, and explicit object-centric decomposition: structured prompting significantly hurts; general and targeted LoRA improve aggregate accuracy but do not remove the persistent orientation weakness; hard negatives improve targeted relational coherence without a significant orientation-accuracy gain; and the tested visual-side LoRA configurations do not outperform the language-side control.

As a sensitivity analysis, removing examples flagged by the automated audit raises orientation accuracy (65.7%→78.5% for 7B General LoRA, 66.4%→76.6% for hard-negative LoRA on the strictest 107-example subset) but does not eliminate error (2B structured shows no gain). The exclusion sets are annotator-dependent: two blind human re-audits under the same taxonomy agree strongly with each other on the failure taxonomy (Krippendorff's α=0.81) while the clean/ambiguous flags agree only moderately with the automated audit (κ=0.36 on the 48 shared persistent-failure cases); details are in the supplementary. Because the removed examples were drawn from persistent failures, part of this gain is expected by construction.

A second failure mode emerges from logical consistency. The zero-shot 7B model is consistent on only 36.9% of complementary facing/facing-away pairs. LM-side LoRA substantially improves coherence, and hard-negative training raises facing consistency to 77.7% while facing accuracy remains unchanged at 68.9%. The extra hard-negative gain is not significant in the pooled strict-family comparison, yet accuracy stays unchanged while consistency moves: the two metrics capture different behavior.

Finally, we test cross-dataset transfer. The 7B model is strong on SITE's official spatial-relationship-reasoning category (75.1% raw; 59.2% chance-adjusted accuracy), yet the VSR-trained adapter does not generalize cleanly: it is statistically neutral overall (54.2%→53.8%, p = 0.66) and significantly degrades the official category (75.1%→71.6%, p = 0.004). A pre-specified, non-official orientation-vocabulary slice is much weaker in the aggregate (48.3% raw; 24.0% chance-adjusted) and shows no positive transfer, but—as we detail below—its composition limits what it can establish, and we therefore do not treat SITE as independent confirmation of the VSR orientation construct.

Our contributions are four reusable components of an evaluation protocol:

- A fine-grained diagnostic ladder. Across ten VSR conditions, two differently sized models, and a large zero-shot sparse-MoE model, several spatial families improve substantially while object-relative orientation remains unusually resistant. The ladder—scaling, prompting, language-side adaptation, targeted data construction, visual-side adaptation, representation probing, and explicit decomposition—is defined operationally so it transfers to other benchmarks and relation families.
- A robustness and sensitivity audit. A label-sensitivity analysis (automated labels with two independent human re-audits), frozen/object-grounded probes, visual-side adaptation, and object-centric decomposition constrain purely annotation-based or single-component explanations. The audit practice (flagged-example exclusion sets, decodability checks against majority baselines) applies to any benchmark whose labels or examples may be ambiguous.
- An accuracy–coherence separation protocol. Complementary-relation tests on VSR's strict relation pairs show that, for the tested conditions, relational consistency can change substantially while task accuracy remains fixed. The protocol

measures consistency against verdict-pattern base rates, so it can be applied wherever complementary formulations of the same relation exist.
- A cross-dataset transfer protocol. SITE shows strong official spatial-relationship performance, while VSR-trained LoRA transfers poorly and significantly harms the official SITE category; the orientation-vocabulary slice is analyzed as exploratory evidence. The protocol (pre-specified subsets, chance-adjusted accuracy, paired transfer tests) is model- and benchmark-agnostic and is reported with negative as well as positive transfer.

The protocol is deliberately modular: each component can be applied independently to other capabilities, benchmarks, and model families. The supplementary specifies the prompts, subsets, decision rules, and released artifacts needed to reproduce the audit.

2. Related Work

Benchmarks for grounded spatial reasoning. VSR introduced a controlled natural-image benchmark with 66 spatial relations and identified facing and parallel as difficult relations for the VLMs studied at the time [7]. What'sUp controls object identity while varying spatial configuration and reports a substantial human–model gap [6]; GSR-Bench broadens grounded evaluation across models and scales [9]. SITE extends the scope to single-image, multi-image, and video settings and organizes tasks by dimensions such as orientation, intrinsic/extrinsic relations, and static/dynamic reasoning [15]. SpatialScore and Mind the Gap further show that spatial competence remains fragmented [10, 16]. We use VSR not to re-establish that spatial reasoning is hard, but to turn its orientation signal into a controlled diagnostic target and use SITE to test transfer.

Diagnosing and improving spatial failures. Several lines of work move beyond aggregate accuracy. AdaptVis relates successful VSR reasoning to attention on relevant object locations [2], while CREG shows that attribution evidence in Qwen2-VL can reflect image layout rather than the queried relation [11]. When More Is Less finds that spatial cues, commonsense information, and reasoning instructions can help or hurt depending on the model and setup [1]. Other work probes visual-concept decodability [13], separates perception from reasoning in spatial planning [17], or supplies geometric inputs for 3D spatial understanding [3]. These studies motivate our combination of frozen/readout diagnostics with adaptation and object-centric decomposition, but do not provide the same end-to-end audit of one relation family across accuracy, coherence, label sensitivity, and transfer.

Taken together, prior work establishes the problem and offers several useful diagnostic lenses, but leaves a methodological gap: benchmark accuracy, representation access, logical consistency, annotation sensitivity, and transfer are usually reported separately. We close that gap with a single controlled diagnostic ladder, with analysis choices and external-evaluation subsets frozen where specified, applied to the same relation family. Our claim is consequently not to discover spatial difficulty, but to make a persistent orientation weakness measurable, falsifiable, and reusable as an evaluation protocol.

Adaptation, specialization, and transfer. Parameter-efficient adaptation such as LoRA [5] can improve a benchmark without improving the underlying capability. We therefore compare general, targeted, hard-negative, projector, and vision-plus-projector updates, then evaluate the strongest language-side adapter on an untouched benchmark. This design distinguishes within-benchmark specialization from transfer: VSR adaptation is neutral overall on SITE, does not improve the orientation subset, and significantly degrades SITE's official spatial-relationship category. Our emphasis on negative transfer complements prior benchmark work by treating transfer failure as an evaluation result rather than an omitted outcome.

Representation probing and logical consistency. Probes can test whether a signal is accessible from frozen representations, but probe capacity can itself learn the task, requiring control baselines and cautious interpretation [4]. Consistency-oriented objectives such as SAGE use geometric and linguistic dualities to train spatial reasoning [8]. We instead use probing and complementary-pair consistency as independent diagnostics: the former asks what the tested readouts can decode, while the latter asks whether answers to logically linked statements agree, so accuracy and coherence can diverge as diagnostic axes.

3. Experimental Setup

3.1. Benchmarks

VSR. We use VSR [7] as the primary diagnostic benchmark. The random-split test set contains 2,195 examples. The benchmark defines 64 relation labels; 61 appear in the evaluated test split. The orientation family contains facing, facing away from, parallel to, and perpendicular to, totaling 137 test examples. All main VSR results use the same held-out test set and deterministic True/False parsing.

SITE. We use SITE [15] for cross-dataset transfer evaluation only and do not train on SITE. The image evaluation contains 2,591 examples: 1,368 single-image and 1,223 multi-image examples. Our primary SITE subset is the official spatial-relationship-reasoning category ($n = 993$). We additionally report a transparently defined, non-official keyword-derived orientation subset ($n = 2{,}024$), analyzed as exploratory evidence rather than as direct replication of the VSR construct. SITE results use the benchmark's native multiple-choice format and chance-adjusted accuracy (CAA) in addition to raw accuracy.

3.2. Models and Conditions

Our compact model is SmolVLM2-2.2B-Instruct, and our larger controlled-study model is Qwen2-VL-7B-Instruct [14]. We retain Qwen2-VL-7B as the controlled anchor because it supports the complete local ladder—the same base model across prompting, LoRA, visual-side adaptation, probing, and decomposition—rather than mixing intervention effects with an uncontrolled architecture change. We evaluate ten canonical VSR conditions: 2B zero-shot, 2B structured prompting, 2B General LoRA, 2B Targeted LoRA, 7B zero-shot, 7B General LoRA, 7B Targeted LoRA, 7B Hard-Neg LoRA, 7B Projector LoRA, and 7B Vision+Projector LoRA. LoRA [5] is used for all parameter-efficient adaptation conditions. We additionally evaluate a large zero-shot sparse-MoE model, MiMo-V2.5 [12] (310B total / 15B activated parameters, dedicated 729M-parameter vision encoder, 2026), served through its public API with the identical frozen prompt, thinking disabled, and greedy decoding (App. A). Its role is external validation of the diagnosed pattern; it is not fine-tuned.

General LoRA uses a proportional sample across relation families. Targeted LoRA allocates most training examples to weak orientation/depth/horizontal families with rehearsal from the remaining relations. Hard-negative LoRA introduces audited complementary orientation examples, primarily facing/facing-away inversions. Vision-side conditions adapt either the multimodal projector or upper vision blocks together with the projector.

3.3. Metrics and Statistical Tests

For VSR we report accuracy overall and by spatial family. For SITE we report raw accuracy and CAA. Paired condition comparisons use McNemar tests when applicable. Clean-label orientation analysis reports the full orientation set and progressively stricter audited subsets. Logical consistency is evaluated by constructing complementary statements for strict relation pairs (left/right, front/behind, facing/facing-away) and testing whether the model assigns opposite verdicts as required. Parallel/perpendicular is treated separately as a soft complement because both relations can be false for oblique objects.

4. Main Results

4.1. Orientation Remains Weak Across Two Differently Sized VLMs

Table 1 shows the central pattern. Moving from the 2B model to the 7B model lifts zero-shot overall accuracy by 6.9 pp, horizontal by 14.6 pp, depth by 6.3 pp, and containment by 5.9 pp, but orientation by only 0.7 pp. General 7B LoRA further raises overall accuracy to 84.7%, yet orientation reaches only 65.7%. Across trained conditions other than structured prompting, orientation remains in a narrow 62–66% band. The large sparse-MoE zero-shot model (MiMo-V2.5, evaluated identically, App. A) falls in the same band: 79.5% overall and 65.7% orientation, with the orientation family again its weakest relation family (per-family table in the supplementary). The relation-specific pattern is therefore not an artifact of the 2B/7B scale range. The MiMo-V2.5 row is also robust to output-format parsing: all 137 orientation outputs were parseable (orientation unchanged at 65.7% parseable-only), and parseable-only overall accuracy is 82.1% vs. 79.5% with the 69 non-conforming outputs counted wrong (App. G).

To quantify the family-relative claim directly, a paired bootstrap over the 2,195 shared examples (10,000 resamples) confirms the horizontal-vs-orientation scaling contrast is robust (+13.9 pp; 95% CI [+2.9, +24.8]; positive in 99.3% of resamples), with depth (+5.5 pp) and containment (+5.1 pp) directionally positive but not individually significant (App. D).

Structured prompting is a negative result: overall accuracy falls from 74.0% to 68.3% at 2B, while orientation falls from 62.8% to 53.3%. A paired exact McNemar test on the full test set confirms that the structured prompt is worse than the minimal baseline ($p \approx 2.2 \times 10^{-7}$). Consistent with recent analyses of information injection on VSR, where additional spatial cues or reasoning instructions can help or hurt depending on the model and setup [1], this result is specific to the tested 2B condition and prompt; it does not establish that prompting generally fails.

4.2. Clean-Label Sensitivity Analysis

An automated audit (LLM-generated labels; the frozen v1 table) identified ambiguous or questionable orientation examples, so we repeat evaluation on progressively stricter subsets as a sensitivity analysis (full table in

Table 1. Canonical VSR test accuracy (%). Orientation improves little with model size or adaptation compared with several other relation families.

Condition	Overall	Orientation	Depth	Horizontal	Containment	Topology/contact
2B zero-shot	74.0	62.8	68.9	70.2	83.4	80.5
2B structured	68.3	53.3	64.9	66.7	78.7	71.7
2B General LoRA	76.6	62.0	71.1	74.3	87.6	81.0
2B Targeted LoRA	76.5	64.2	70.8	75.1	87.6	81.4
7B zero-shot	80.9	63.5	75.2	84.8	89.3	80.3
7B General LoRA	84.7	65.7	82.3	87.4	92.9	84.4
7B Targeted LoRA	83.9	64.2	81.7	87.1	91.7	85.3
7B Hard-Neg LoRA	84.3	66.4	80.7	87.1	89.9	84.6
7B Projector LoRA	82.9	64.2	78.6	87.1	89.3	85.3
7B Vision+Projector LoRA	83.1	64.2	78.0	87.1	89.9	84.4
MiMo-V2.5 zero-shot	79.5	65.7	74.8	81.9	87.1	84.6

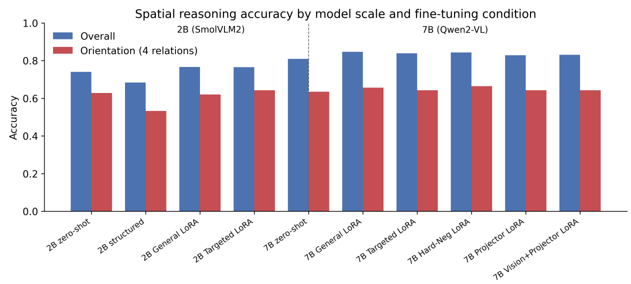

Figure 1. Overall versus orientation performance across the two VLMs and adaptation conditions.

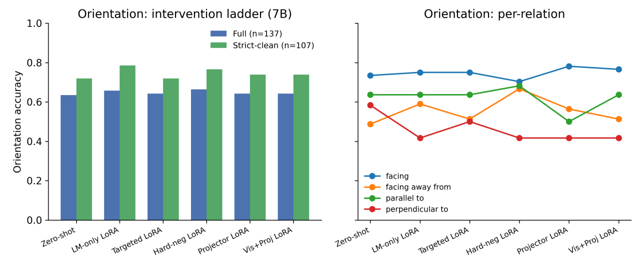

Figure 2. Orientation performance across 7B interventions, including clean-label analysis and per-relation behavior.

the supplementary). The best full-set orientation result is 66.4%; on the automated strictest 107-example subset, the best result rises to 78.5% for 7B General LoRA. Because auditing changes the evaluated sample, these are not matched comparisons against other relation families. Two blind human re-audits under the same taxonomy (binary flag on all 137, eight-class taxonomy on the 48 persistent failures) yield three distinct analyses, reported separately.

Taxonomy-derived strict masks. Each rater's strict mask raises orientation accuracy without eliminating error—automated 78.5% (n=107), human A 77.6% (n=107), human B 78.8% (n=104)—and 2B structured prompting still shows no gain under any of the three masks (53.3%, 51.4%, 51.0%).

Binary clean/ambiguous masks. The two humans agree substantially on the 137-example binary flag (κ=0.645, 82.5% agreement) but only moderately with the automated audit (κ=0.36 on the 48 cases rated by both sources; the automated audit never flagged the remaining 89), and the taxonomy agrees only weakly with the automated labels (α=0.14). Under the human binary masks the clean-subset gains largely vanish (best 66.7% and 66.2% vs. the 66.4% full-set best), tying the gains to the taxonomy-derived masks rather than to the binary flag. Audit reliability is itself model-dependent: two further LLM auditors (GPT-5.6 Sol and Gemini Spark) agree with the humans only weakly (κ=0.34 and κ=0.03 on the 137-example flag; Gemini flags 126/137 as clean), so LLM-generated ambiguity labels are sensitivity tools, not authoritative ground truth (App. F).

Consensus analysis. Under the consensus union mask (n=62), the trained conditions again improve (7B General LoRA 77.4%) but 7B zero-shot does not (58.1%), and the 2B structured no-gain pattern does not survive this mask (69.3%), consistent with annotator dependence.

The robust conclusion is narrower: taxonomy-derived strict masks raise orientation scores but do not eliminate error across conditions; the audit is exploratory qualitative evidence, not a ruling on label noise (frozen v1 and both human tables in the supplementary).

5. Why Does Orientation Remain Difficult?

5.1. Frozen Representation Probes

We probe frozen Qwen2-VL-7B-Instruct visual representations (no fine-tuning). Mean-pooled ViT and post-merger features support only weak relation decoding for facing versus facing-away, parallel versus perpendicular, and a four-way orientation task. Linear and MLP probes generally remain near majority baselines; preserving patch-level structure yields a suggestive point estimate for parallel/perpendicular (patch-vote 73.5% vs. 64.7% majority), but that test set is small (n=34) and the confidence interval overlaps the majority baseline.

To reduce the confound that global features might infer relations without identifying the relevant objects, we additionally extract subject/reference regions and construct grounded visual features from subject, object, difference, and elementwise-product representations. Object-box conditioning with the model's own grounding predictions and mean-pooled region features does not recover robust orientation decodability. Following standard cautions about diagnostic classifiers [4], we phrase these results as decodability under the tested readouts, not as proof that the information is absent from the network. Linear decodability of visual concepts has been probed in driving-oriented VLMs [13], and directional information in this model family has been inspected via attribution [11]; our probe analysis is not the first of either kind. It differs by targeting VSR's orientation family with pooled, patch-vote, and object-grounded readouts over the exact conditioning features of the generative model, paired with adaptation and consistency interventions. Detailed probe tables and plots are in the supplementary.

5.2. Visual-Side Adaptation and Object-Centric Decomposition

If the frozen representation were the sole limiting factor, directly adapting visual or projector components might recover orientation performance. For the trained checkpoints evaluated here, it did not: projector LoRA and vision+projector LoRA both produced 64.2% orientation accuracy, below the 65.7% LM-only control, while overall accuracy was nominally lower (82.9% and 83.1% vs. 84.7%; unadjusted p=0.0043/0.0122; these effects do not survive a blanket multiple-comparison adjustment and are not load-bearing, App. D). This suggests that the tested low-rank adaptation of the visual pathway does not close the orientation gap.

We also test a box-based two-stage decomposition. Its four-way relation module classifies facing/facing-away/parallel/perpendicular against ground-truth relation labels at 44–46% cross-validation accuracy (49.9% majority baseline; uniform four-class chance 25%), so the tested relation classifier does not outperform the empirical majority baseline. Consistent with that, the pipeline's True/False verdicts match the end-to-end control's correctness on only 55.1–58.3% of the 127 boxed orientation statements (per-relation agreement 25.0–69.4%, weakest for parallel; App. D); in the committed analysis (10 no-box statements scored wrong) the verdicts diverge from the control on 61–65 of the 137 statements, with the control correct and the pipeline wrong in 44–47 of them. These are agreement scores, not ground-truth accuracies: the committed script scored the pipeline against the control's verdicts, so we report the tested decomposition as an agreement diagnostic that fails to reproduce the generative model's behavior, not as a direct accuracy comparison (per-example ground-truth scoring is provided by the released script and requires the frozen embeddings, App. G). Two-dimensional box geometry is structurally insufficient for intrinsic facing direction, and the tested region representations do not supply enough signal to close the gap.

5.3. Logical Consistency Reveals a Distinct Failure Mode

We construct complementary statements for strict relation families. A coherent model should return opposite truth values for left/right, front/behind, and facing/facing-away formulations of the same object pair. Complementary-pair and transformation-based consistency evaluation has prior literature, including consistency-oriented training [8]; our contribution is the VSR-specific diagnostic below, built on strict complementary relation pairs with observed verdict-pattern and base-rate analysis. The zero-shot 7B model is consistent on only 58.0% of left/right pairs, 57.0% of front/behind pairs, and 36.9% of facing pairs. The large sparse-MoE zero-shot model does not repair coherence either: it reaches 42.7% facing-pair consistency under the same definition (vs. 36.9% for the 7B zero-shot model; 44.0% on the 100 pairs with parseable verdicts, three excluded for non-conforming outputs), with the same False-defaulting verdict pattern as the smaller models (0 both-True, 56.0% both-False; verdict table in the supplementary). Scale alone therefore does not resolve the coherence failure mode. The observed verdict patterns show what drives this: a False-defaulting bias that yields both-False verdicts on 63.1% of pairs and no both-True verdicts (0/103), with expected consistency under verdict independence at 30.2%, below the observed 36.9% (verdict table in the supplementary): the dominant phenomenon is a refusal

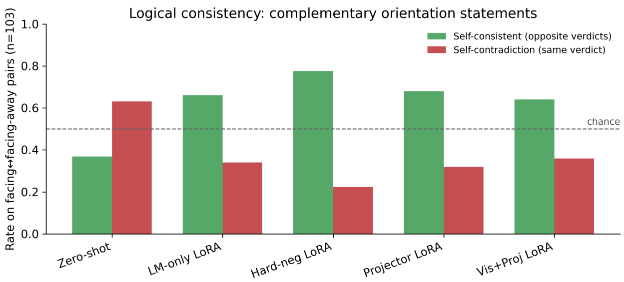

Figure 3. Self-consistency on complementary facing/facing-away relations across 7B conditions.

to affirm either proposition, rather than a preference for contradictory affirmations.

LM-only LoRA raises facing consistency to 66.0% and front/behind to 70.7% (pooled strict-family significantly better than zero-shot, $p < 0.0001$); hard-negative training raises facing consistency to 77.7% with facing accuracy identical at 68.9% for both, although the additional pooled consistency effect over LM-only is not significant ($p = 0.29$). The key result is separability: identical facing accuracy coexists with substantially different facing consistency; we do not claim a general coherence algorithm. Training also changes the failure mode rather than eliminating it: both-False facing verdicts drop from 63.1% to 9.7% (LM-only) while both-True contradictions rise from 0% to 24.3%, a disclosed specialization trade-off.

6. Cross-Dataset Transfer to SITE

We evaluate the frozen 7B base model and the VSR-trained 7B General LoRA adapter on the same 2,591 SITE image examples using SITE's native multiple-choice protocol [15], with no SITE training. Because the pre-specified decision rule resolved to branch 2, this adapter run is a post-protocol transfer check rather than a pre-specified step (protocol and outcome in the supplementary). The model is strong on the official spatial-relationship-reasoning category: 75.1% raw accuracy and 59.2% CAA. Applying the VSR-trained adapter does not transfer cleanly: it is statistically neutral overall (54.2%→53.8%, $p = 0.66$) and significantly degrades the official category (75.1%→71.6%, $p = 0.004$; 52 examples fixed vs. 87 broken). VSR gains therefore transfer poorly and can induce negative transfer; we do not equate benchmark-specific fine-tuning gains with general spatial-reasoning improvement.

The pre-specified, non-official orientation-vocabulary slice is much weaker in the aggregate (48.3% raw / 24.0% CAA vs. 75.1% / 59.2% for the official category) and shows no positive transfer under

Table 2. SITE cross-dataset transfer (images). Entries are raw accuracy / CAA (%).

Subset	Zero-shot	VSR-LoRA	Δ
All images	54.2 / 31.1	53.8 / 30.5	−0.4
Official spatial-rel.	75.1 / 59.2	71.6 / 53.4	−3.5
Orient. heuristic	48.3 / 24.0	48.7 / 24.5	+0.4

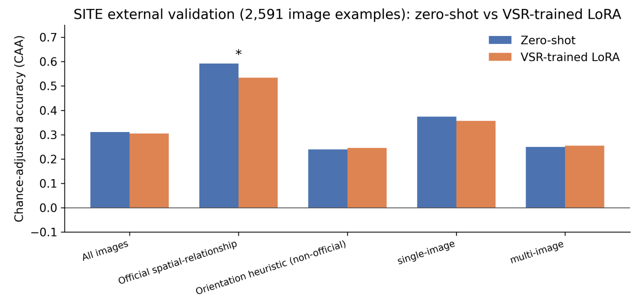

Figure 4. SITE CAA for zero-shot and VSR-trained LoRA. The official spatial-relationship category degrades significantly under transfer.

the adapter (48.3%→48.7%, $p = 0.70$). Because the slice is keyword-derived and heterogeneous (43% of its examples come from multi-view/cross-image reasoning, only 24% from the spatial-relationship category), we treat it as exploratory evidence rather than replication of the VSR construct. A composition-confound analysis of the 2,591 zero-shot predictions (supplementary §E) supports this caution: after controlling for official category, source dataset, modality, and number of options, the orientation flag is not significantly associated with correctness (odds ratio 0.84, 95% CI $[0.60, 1.16]$, $p = 0.29$), although it remains descriptively 7.5pp lower within the official spatial-relationship category itself (71.3% vs. 78.8%, n=488/505). A post-hoc high-precision subset restricted to VSR-construct terms (facing, facing away, parallel, perpendicular; n=177) yields 44.1% raw (95% CI $[37.0, 51.4]$; CAA −2.3%) and is reported as exploratory only. We therefore do not treat SITE as independent confirmation of the VSR orientation bottleneck; the transfer results stand on the official category, where the model is strong and the adapter demonstrably does harm.

7. Discussion

Orientation is not spatial reasoning broadly. The strongest conclusion is deliberately narrow. The 7B model is strong on several VSR families and SITE's official spatial-relationship category. The persistent weakness concerns the object-relative orientation family in VSR; on SITE, the orientation-vocabulary slice is analyzed as exploratory evidence rather than as independent confirmation. This framing is consistent with the original VSR observation [7] while extending it to modern generative VLMs and a broader intervention analysis.

Decodability and coherence both matter, as diagnostics. The probes show that orientation is not robustly decodable under the tested frozen-feature readouts, while consistency experiments expose large logical failures even for relations answered above chance. Neither result alone identifies a complete causal mechanism, and both are diagnostic rather than causal-localization claims. Together they support a mixed account in which orientation information is difficult to access reliably under the tested readouts and relational decisions are not always propagated coherently across complementary formulations.

Task adaptation can specialize rather than generalize. General LoRA improves VSR overall accuracy substantially, but its SITE transfer is neutral overall and significantly negative on the official subset. Hard negatives improve a targeted coherence behavior with relation-specific tradeoffs. Benchmark-specific fine-tuning gains should therefore not be equated with general spatial-reasoning improvement.

8. Limitations

Statistical power is limited at the scale of the orientation diagnostic. The VSR orientation family contains only 137 test examples, and the smallest per-relation subset, perpendicular, has n=12; perpendicular results are therefore reported descriptively throughout. The Wilson 95% confidence intervals on the family-level point estimates are correspondingly wide—62.8% $[54, 70]$, 63.5% $[55, 71]$, 65.7% $[57, 73]$, and 66.4% $[58, 74]$ for 2B zero-shot, 7B zero-shot, 7B General LoRA, and 7B Hard-Neg LoRA (n=137 each)—so the trained conditions' 62–66% band should be read as a set of statistically overlapping point estimates rather than a fine-grained ranking. Several smaller point estimates have confidence intervals that include the majority baseline or chance: the patch-vote parallel/perpendicular probe (73.5%, n=34; 95% CI $[57, 85]$ vs. the 64.7% majority), the perpendicular subset for the trained 7B conditions (41.7%, n=12; $[19, 68]$ vs. 50%), and the 7B zero-shot facing-away subset (48.7%, n=39; $[34, 64]$). Paired McNemar tests on the orientation family rest on small discordant-pair counts and have limited power: the non-significant projector and vision+projector orientation comparisons (p=0.86, p=0.85) indicate an absence of detectable effect, not evidence of equivalence. Where the paper draws conclusions, it relies on the full-set paired tests and the Holm-adjusted battery in the supplementary rather than on these small-n point estimates alone.

The clean-label audit's exclusion sets originate from an automated LLM audit: two independent human re-audits reached only moderate per-item agreement with it (κ=0.36 on the clean/ambiguous flag; Krippendorff's α=0.14 on the eight-class taxonomy) while agreeing strongly with each other on the failure taxonomy (α=0.81); both reproduced the qualitative conclusions under their own taxonomy-derived strict masks, so the sets are annotator-dependent, and the frozen table is reported as v1 with both human re-audit tables alongside. The SITE orientation slice is keyword-derived rather than an official benchmark category; a composition-confound analysis shows its aggregate score is largely explained by task composition, so we do not treat SITE as independent confirmation of the VSR orientation bottleneck. Our probing methods test accessible decodability under specific linear/MLP and pooling choices; failure to decode does not prove information is absent. Visual-side experiments use LoRA rather than full end-to-end retraining. We study two differently sized VLMs from different families plus a large sparse-MoE third model; this is a cross-model comparison, not a controlled scale experiment. All trained conditions are single-checkpoint runs: the paired McNemar tests quantify disagreement on the fixed test examples, not training-run variability, and these single-seed results do not estimate it (a multi-seed extension is prepared in scripts/run_seed_variance.py). The MiMo-V2.5 evaluation is served through a public API: 3.1% of its outputs did not conform to the one-word format and are counted as wrong (invalid outputs are never guessed), and 294 responses contained chain-of-thought text alongside the verdict; both are disclosed in the committed predictions (App. A). The MiMo-V2.5 row was evaluated on 2026-08-12 through the OpenCode Go gateway; API-served models can be updated without notice, so this row's reproducibility depends on the serving snapshot (evaluation date, model id, and per-request token accounting are recorded in results/mimo/). The SITE evaluation is image-only; SITE video reasoning remains future work. Finally, several interventions are evaluated as negative results, so lack of improvement should be interpreted for the tested configurations rather than as impossibility claims about all prompting, adaptation, or architectural methods.

WACV #*****

9. Conclusion

Aggregate spatial accuracy hides a persistent orientation bottleneck in the studied VLMs: a larger VLM and standard adaptation substantially improve several spatial families while leaving object-relative orientation resistant, and a large 2026 sparse-MoE model reproduces the pattern. Auditing, probing, visual-side adaptation, and decomposition constrain simple explanations; complementary-relation tests show accuracy and coherence are distinct axes that both remain weak in the large model. SITE transfer shows VSR-specific adaptation can negatively transfer. Evaluation should report relation-specific performance, transfer, and logical coherence rather than aggregate accuracy.

References

[1] Muku Akasaka and Soyeon Caren Han. When more is less: A systematic analysis of spatial and commonsense information for visual spatial reasoning. arXiv preprint arXiv:2602.21619, 2026. 3, 4

[2] Shiqi Chen, Tongyao Zhu, Ruochen Zhou, Jinghan Zhang, Siyang Gao, Juan Carlos Niebles, Mor Geva, Junxian He, Jiajun Wu, and Manling Li. Why is spatial reasoning hard for vlms? an attention mechanism perspective on focus areas. In Proceedings of the 42nd International Conference on Machine Learning (ICML), 2025. 3

[3] Erik Daxberger, Nina Wenzel, David Griffiths, Haiming Gang, Justin Lazarow, Gefen Kohavi, Kai Kang, Marcin Eichner, Yinfei Yang, Afshin Dehghan, and Peter Grasch. MM-Spatial: Exploring 3d spatial understanding in multimodal LLMs. In Proceedings of the IEEE/CVF International Conference on Computer Vision, pages 7395–7408, 2025. 3

[4] John Hewitt and Percy Liang. Designing and interpreting probes with control tasks. In Proceedings of the 2019 Conference on Empirical Methods in Natural Language Processing and the 9th International Joint Conference on Natural Language Processing, pages 2733–2743. Association for Computational Linguistics, 2019. 3, 6

[5] Edward J. Hu, Yelong Shen, Phillip Wallis, Zeyuan Allen-Zhu, Yuanzhi Li, Shean Wang, Lu Wang, and Weizhu Chen. LoRA: Low-rank adaptation of large language models. arXiv preprint arXiv:2106.09685, 2021. 3, 4

[6] Amita Kamath, Jack Hessel, and Kai-Wei Chang. What's "up" with vision-language models? investigating their struggle with spatial reasoning. arXiv preprint arXiv:2310.19785, 2023. 1, 3

[7] Fangyu Liu, Guy Emerson, and Nigel Collier. Visual spatial reasoning. Transactions of the Association for Computational Linguistics, 11:635–651, 2023. 1, 3, 8

[8] Junming Liu, Yuqi Li, Yifei Sun, Maonan Wang, Piotr Koniusz, Yirong Chen, and Ding Wang. Self-evolving spatial reasoning in vision language models via geometric logic consistency. arXiv preprint arXiv:2605.18162, 2026. 3, 6

[9] Navid Rajabi and Jana Kosecka. GSR-BENCH: A benchmark for grounded spatial reasoning evaluation via multimodal LLMs. arXiv preprint arXiv:2406.13246, 2024. 1, 3

[10] Ilias Stogiannidis, Steven McDonagh, and Sotirios A. Tsaftaris. Mind the gap: Benchmarking spatial reasoning in vision-language models. arXiv preprint arXiv:2503.19707, 2025. 3

[11] Kaizhen Tan, Yang Feng, and Heqing Du. CREG: Compass relational evidence graph for characterizing directional structure in VLM spatial-reasoning attribution. arXiv preprint arXiv:2603.20475, 2026. 3, 6

[12] Xiaomi MiMo Team. Mimo-v2.5. https://huggingface.co/collections/XiaomiMiMo/mimo-v25, 2026. 4

[13] Nikos Theodoridis, Reenu Mohandas, Ganesh Sistu, Anthony Scanlan, Ciarán Eising, and Tim Brophy. Probing visual concepts in lightweight vision-language models for automated driving. Transactions on Machine Learning Research, 2026. 3, 6

[14] Peng Wang, Shuai Bai, Sinan Tan, Shijie Wang, Zhihao Fan, Jinze Bai, Keqin Chen, Xuejing Liu, Jialin Wang, Wenbin Ge, Yang Fan, Kai Dang, Mengfei Du, Xuancheng Ren, Rui Men, Dayiheng Liu, Chang Zhou, Jingren Zhou, and Junyang Lin. Qwen2-VL: Enhancing vision-language model's perception of the world at any resolution. arXiv preprint arXiv:2409.12191, 2024. 4

[15] Wenqi Wang, Reuben Tan, Pengyue Zhu, Jianwei Yang, Zhengyuan Yang, Lijuan Wang, Andrey Kolobov, Jianfeng Gao, and Boqing Gong. SITE: towards spatial intelligence thorough evaluation. In Proceedings of the IEEE/CVF International Conference on Computer Vision, pages 9058–9069, 2025. 1, 3, 4, 7

[16] Haoning Wu, Xiao Huang, Yaohui Chen, Ya Zhang, Yanfeng Wang, and Weidi Xie. Spatialscore: Towards comprehensive evaluation for spatial intelligence. In Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition, pages 31029–31041, 2026. 3

[17] Qiucheng Wu, Handong Zhao, Michael Saxon, Trung Bui, William Yang Wang, Yang Zhang, and Shiyu Chang. VSP: Diagnosing the dual challenges of perception and reasoning in spatial planning tasks for MLLMs. In Proceedings of the IEEE/CVF International Conference on Computer Vision, pages 2270–2280, 2025. 3